

INTENSITY PREDICTION MODEL FOR TROPICAL CYCLONE RAPID INTENSIFICATION EVENTS

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ABSTRACT

Tropical Cyclones (TC) create strong wind and rain and can cause significant human and financial losses. Many major hurricanes in the Atlantic Ocean undergo rapid intensification (RI). RI events happen when the strength of the storm increases rapidly within 24 hours. Improving the hurricane's intensity prediction model by accurately detecting the occurrence and predicting the intensity of an RI event can help avoid human and financial losses. In this paper we analyzed RI events in the Atlantic Basin and investigated the use of a combination of three different models to predict RI events. The first and second models are simple location and time-based models which use the conditional probability of intensification given the day of the year and the location of the storm. The third model uses a data mining technique which is based on the Extensible Markov Chain Model (EMM) which clusters the hurricane's life cycle into states and then uses the transition probabilities between these states for prediction. One of the main characteristics of a Markov Chain is that the next state depends only on the current state and by knowing the current state we can predict future states which provide us with estimates for future intensities. In future research we plan to test each model independently and combinations of the models by comparing them to the current best prediction models.

KEYWORDS

Tropical Cyclone, Rapid Intensification, Data Mining, Clustering, Bayesian Inference.

1. INTRODUCTION

Different statistical and dynamical models were developed to predict Tropical Cyclone (TC) tracks and intensities. Current track models are much more accurate compared to intensity models (Jovanovic, et al., 2011). One reason is the existence of rapid intensification (RI) events which are defined in Kaplan, et al., (2010) as events in the life cycle of a TC where the maximum sustained wind speed increases by 30 knots or greater within 24 hours. The number of RI events is small compared to the total number of storms that happened over the years; however their potential impact is large since they represent a significant threat. RI events appear between June and November and their main causes are still not clear which makes them hard to predict. A rapid intensification index (RII) was developed in Kaplan, et al., (2010). It is a regression-based statistical model used to forecast RI events based on predictors from the SHIPS hurricane intensity prediction model (DeMaria & Kaplan, 1994). A study done by Wang & Zhou, (2008) shows how often RI events occur in different locations based on seasonal, intra-seasonal, and inter-annual time scales. Another study done by DeMaria & Kaplan, (2003) shows that there is a great potential of RI when a storm meets deep layers of

warm water. However, RII's probability of detecting RI events is only between 15% and 59% and requires improvement.

Our proposed model determines whether an RI event will take place within 24 hours and predicts its intensity. By analyzing RI events, we found that the location and day of the year are important factors that help predict those events. Therefore, we have built three different models. The time-based model counts the number of RI events and storms for each day, and then smoothes the values of RI events and storms to determine the potential of RI given the day of the year. The location based model divides the map into equal-sized grid and estimates the potential of RI for each point on the grid by counting the number of RI events and storms that are within a distance of m miles from each point. This gives the potential of RI given the location and path of the storm. The state based model uses the storms' historical data from 1982-2011 and uses the same predictors used by SHIPS and RII. It builds a graph that presents the life cycle of each storm using the Extensible Markov Chain Model (EMM).

2. RAPID INTENSIFICATION

To perform our analysis we analyzed every storm from 1982-2011 and extracted the RI cases where the change in intensity is equal to or greater than 30 knots within 24 hours. Then we visualized the RI events. Figure 1 summarizes the location of all RI cases in the dataset. Figure 2 shows how the intensity of an example storm with an RI event changes over time. We also plot RI events based on the month and found that the behavior, location, and number of events changed from one month to another.

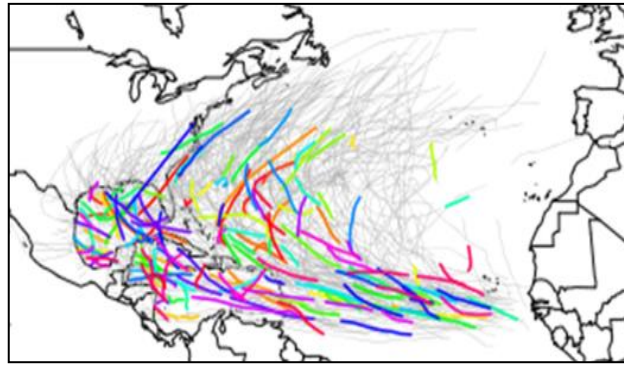


Figure 1. RI events that happened between 1982 and 2011 in the Atlantic Ocean.

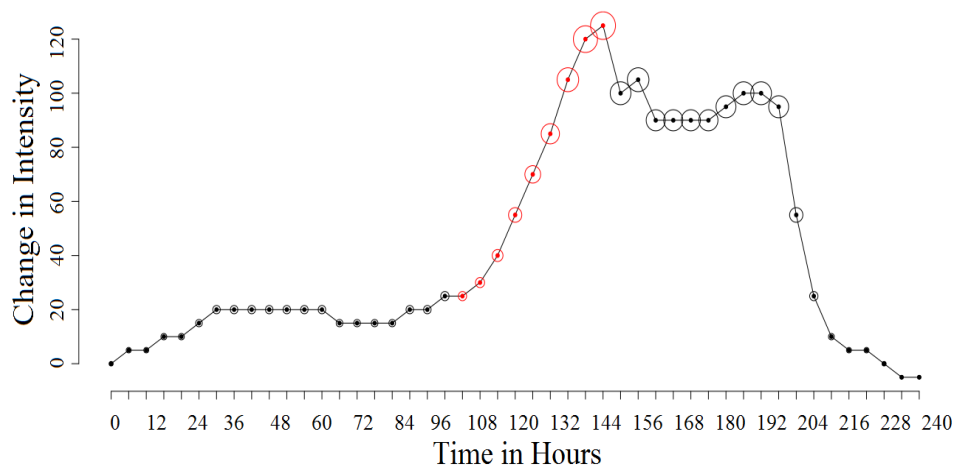


Figure 2. An example of the change in intensity over time of a TC that underwent RI. The size of each circle is the intensity at a certain time.

3. TIME BASED MODEL

For all models we use a training set that includes historical storm data from 1982 to 2010. In the hurricane data we have observed that the frequency of RI's depends on the day of the year. Therefore, we built a model that is based on which day of the year a storm occurs. To build our model, we first counted the number of RI events that occurred on each day of the year using the training set. The count data has high variability from day to day. Therefore, we decided to smooth the values by averaging the counts using 15, 30, and 60 day windows (shown in Figure 3). Next, we repeated the same process by counting the total number of storms that happened on each day of the year then smoothed the values as shown on Figure 3. Finally, we applied Bayes' Theorem to calculate an estimate for the conditional probability of seeing a RI given an active hurricane on a given day (H_{day}):

$$\hat{P}(RI|H_{day}) = \frac{P(H_{day}|RI)P(RI)}{P(H_{day})}$$

The probability of an RI event is calculated by dividing the number of RI storms by the number of RI and non-RI storms as follows:

$$P(RI) = \frac{\sum RI}{\sum (RI + non - RI)}$$

The probability that a storm will take place in a certain day is calculated by dividing the number of storms on that day by the number of RI and non-RI storms as the following:

$$P(H_{day}) = \frac{\sum_{day} (RI + non - RI)}{\sum (RI + non - RI)}$$

The probability of each day given that an RI event has occurred is calculated by counting the number of RI events on that day divided by the total number of RI events as follows:

$$P(H_{day}|RI) = \frac{\sum_{day} RI}{\sum RI}$$

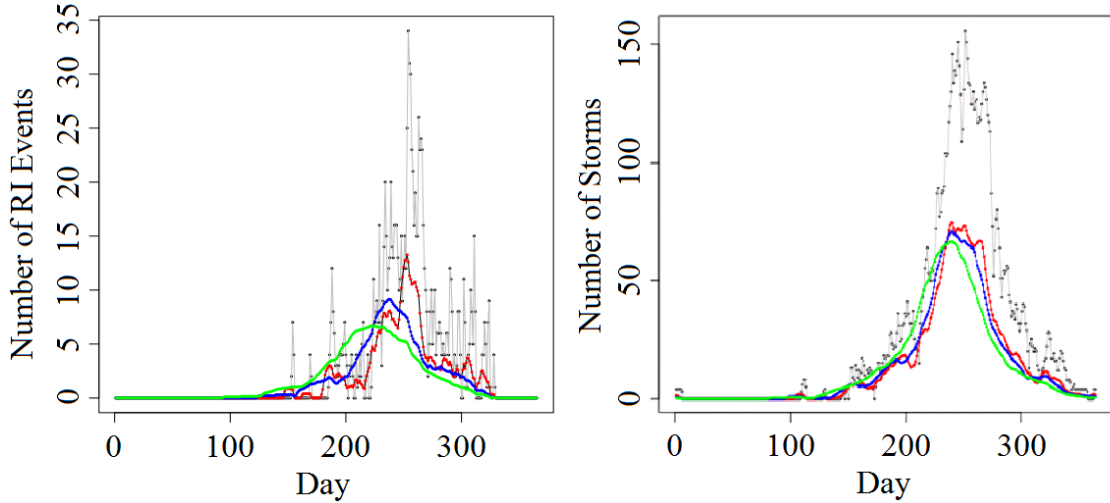


Figure 3. The original number of RI events and storms for each day of the year and their smoothed values. The gray color represents the original values while red, blue, and green represents 15, 30, and 60 day windows.

4. LOCATION BASED MODEL

Since the occurrence of an RI event also depends on the location of the storm, we built a location-based model which estimates the probability of an RI event given its location. To build the model, we first divided

the map into an equal sized grid. Each point on the grid is a spatial coordinate and is represented by its longitude and latitude. Then, for each point we defined a constant which represents a radius in miles. Using the training set we counted the total number of RI events and storms that appeared within the specified radius by measuring the distance between the current point on the grid and all other storms or RI events. Finally, we applied again Bayes' Theorem to calculate the probability that an RI event will occur for a hurricane close to each point on the grid given the location of the storm. Figure 4 shows our results and the following formula was used to find the probabilities:

$$\hat{P}(RI|H_{loc}) = \frac{P(H_{loc}|RI)P(RI)}{P(H_{loc})}$$

The probability of RI is similar to the way it was defined in the time model. The probability that a storm will take place in a certain location is defined as follows:

$$P(H_{loc}) = \frac{\sum_{loc}(RI + non - RI)}{\sum(RI + non - RI)}$$

The probability of each location given that an RI event has occurred is calculated as follows:

$$P(H_{loc}|RI) = \frac{\sum_{loc}(RI)}{\sum RI}$$

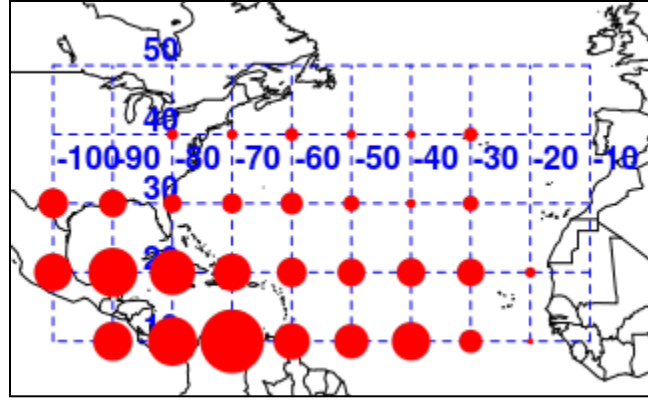


Figure 4. The size of the circles shows the probability of RI for each point on the grid. The numbers shown on the grid are the longitude and latitude degrees that are used to plot the grid.

5. STATE BASED MODEL

Forecasting the intensity of a TC requires a time series model that respects its temporal structure. We suggest an EMM model for representation. It can be used to predict the future intensity based on the current state. Each storm's life cycle is represented with states (clusters) and transitions. Each state represents a combination of several predictors (wind speed, wind direction, sea surface temperature, etc.) that describe the storm at 6 hour time intervals. Transition between states represents the probability that a storm will change from one state to another. The model can be learned directly from the data (see Su, et al., December 2010). Figure 5 shows an example. When a new TC appears, we can predict its future behavior depending on its current state and the transition probabilities in the model. As we mentioned before, RI is a special case that happens during the life cycle of a TC. Using the Extensible Markov Chain (EMM) model, we can find possible RI events by inspecting the intensity of states which are reachable via 6-hour transitions (within 24 hours). If the change in intensity from the current state is equal to or greater than 30 knots, it means that there is a potential of RI. Figure 5 gives an example. The following formula is used to calculate the probability of an RI event:

$$\hat{P}(RI|H, state) = P(RI| state_i \rightarrow state_j \wedge I(state_j) - I(state_i) \geq 30 \wedge i \xrightarrow{24} j)$$

Where $I(state)$ is the average intensity of the state, 30 stands for 30 knots, and the arrow represents a 24 hour transition in the model.

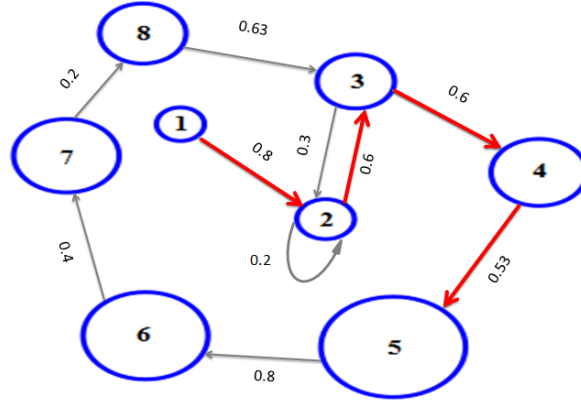


Figure 5: This figure shows an EMM model used to predict RI. The circles are the states, and the size of each circle is its average intensity. The arrows are the transitions and they carry the probability of moving from one state to another (transitions with probabilities of less than .2 are not shown in the graph). State 1 through 5 presents an RI event (the change in intensity from state 1 to state 5 is greater than 30 knots).

6. CONCLUSION

In this paper we have developed two simple and one more complex model for RI prediction. We have implemented the models in R and plan to do real time experiments to compare the models to each other and to the RII developed by Kaplan, et al., (2010). We are also working on combining these and other models using ideas from Bayesian Inference to combine the three models assuming that each model is independent from the other. The uncertainty of the features causing RI events and the small number of RI events compared to the total number of storms makes it a challenging problem especially because their occurrences are a real threat. Understanding the behavior of RI events, choosing the right features, and applying different data mining techniques could improve the prediction of RI events which will improve the prediction of TCs.

REFERENCES

- DeMaria, M. & Kaplan, J., 1994. A statistical hurricane intensity prediction scheme (SHIPS) for the Atlantic basin. *Weather and Forecasting*, 9, pp. 209-220.
- DeMaria, M. & Kaplan, J., 2003. Large-Scale Characteristics of Rapidly Intensifying Tropical Cyclones in the North Atlantic Basin. *Weather and Forecasting*, 18(6), pp. 1093-1108.
- Jovanovic, V., Dunham, M., Hahsler, M. & Su, Y., December 2011. Evaluating hurricane intensity prediction techniques in real time. *Third IEEE ICDM Workshop on Knowledge Discovery from Climate Data, Proceedings of the of the 2011 IEEE International Conference on Data Mining Workshops (ICDMW 2011)*. pp. pages 23-29.
- Kaplan, J., DeMaria, M. & Knaff, J., 2010. A revised tropical cyclone rapid intensification index for the Atlantic and eastern North Pacific basins. *Weather and Forecasting*, 25(1), pp. 220-241.
- Su, Y., Chelluboina, S., Hahsler, M. & Dunham, M. H., December 2010. A new data mining model for hurricane intensity prediction. *Second IEEE ICDM Workshop on Knowledge Discovery from Climate Data: Prediction, Extremes and Impacts, Proceedings of the of the 2010 IEEE International Conference on Data Mining Workshops (ICDMW 2010)*, pp. 98-105.
- Wang, B. & Zhou, X., 2008. Climate variation and prediction of rapid intensification. *Meteorology and Atmospheric Physics*, 99, pp. 1-16.